THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 11: **Keeping Time with the Skies** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 45 degrees	Shadow = height.
2	**(b)** 354	12 x 29.5.
3	**(d)** west after sunset	Young Moon.
4	**(a)** to the south of the observer	Shadow is opposite to the Sun.
5	**(b)** 11 days	365 - 354.
6	**(c)** sunset	Opposite to the Sun.
7	**(b)** 15 days	Half a month.
8	**(c)** 33 days	3 x 11.
9	**(c)** sunrise	Near the Sun.
10	**(a)** 15 January	Half a lunar month.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Shadow = height (1); Angle = 45 degrees (1).	1+1
12	Lunar month is 29.5 days (1); 12 x 29.5 = 354 (1).	1+1
13	Heights are in the ratio of shadows (1); Height = 12 m (1).	1+1
14	It is below the northern horizon (1); Because it lies above the northern axis (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	9 am: long, west (1); 12 noon: shortest (1); 3 pm: long, east and symmetrical with 9 am (1).	1+1+1
16	Gregorian: solar, 365 or 366 days (1); Islamic: lunar, about 354 days (1); Hindu: lunisolar with extra months (1).	1+1+1
17	North hemisphere tilts toward the Sun (1); Sun is above horizon for longer (1); Day is longer in summer (1).	1+1+1
	OR: Mark the shortest shadow (1); It lies along the north-south line (1); Shadow points to the north in India (1).	1+1+1
18	Altitude of the Pole Star equals latitude (1); Measured with an instrument (1); At the equator it is on the horizon (1).	1+1+1
	OR: Festivals by lunar dates (1); Tides caused by the Moon's pull (1); Holi on a full moon, Diwali on a new moon (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Fix a vertical stick (1); Mark the tip of the shadow every 10 minutes (1); Shortest shadow is local noon (1); It lies along the north-south line (1); Compare with clock time (1).	1+1+1+1+1
	OR: Latitude equals the altitude, so 25 degrees North (1); Find the Big Dipper (1); Extend the pointer stars (1); Five times the gap (1); Pole Star found (1).	1+1+1+1+1
20	New moon: Moon near Sun, rises with Sun (1); First quarter: rises at noon (1); Full moon: opposite the Sun, rises at sunset (1); Last quarter: rises at midnight (1); About 29.5 days cycle (1).	1+1+1+1+1
	OR: Sun's noon height changes (1); Highest in summer, so shortest shadow (1); Lowest in winter, so longest shadow (1); Due to the axial tilt (1); Equinox in between (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 11: **Keeping Time with the Skies** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** Adhik maas	Corrects the gap.
2	**(c)** noon	Half-lit and a quarter way round.
3	**(a)** 146097 days (with 97 leap years)	400 x 365 + 97.
4	**(c)** latitude	At the north pole it is overhead.
5	**(c)** full moon	Decreasing after full.
6	**(a)** 2024	Divisible by 4.
7	**(a)** overhead	Directly above.
8	**(b)** a new moon day	Moon between the Sun and the Earth.
9	**(d)** 1900	Century years must be divisible by 400.
10	**(b)** on the northern horizon	Altitude equals 0 degrees.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Seasons arise from the axial tilt (1); Earth is closest to the Sun in early January, which is winter in India (1).	1+1
12	3 x 365 + 366 (1); = 1461 days (1).	1+1
13	Moon moves eastward in its orbit (1); Earth must rotate a little more to bring it up (1).	1+1
14	New moon: Moon between Sun and Earth (1); Full moon: Earth between Sun and Moon (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Phases occur every month; eclipses are rare (1); Phases due to the visible fraction of the lit half; eclipses due to shadow (1); Eclipses need exact alignment (1).	1+1+1
16	Earth moves in its orbit (1); Night side faces different stars (1); Some are daytime stars in some months (1).	1+1+1
17	Explained Earth's rotation (1); Computed the length of the year (1); Developed methods for eclipses (1).	1+1+1
	OR: Axis tilted (1); When north tilts toward the Sun it is summer (1); South tilted away has winter (1).	1+1+1
18	Stars trace circular arcs (1); Centre of arcs is the Pole Star (1); Due to rotation of the Earth (1).	1+1+1
	OR: Day length = sunset - sunrise (1); Longest in June (1); Shortest in December (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Earth's axis tilted (1); Revolution keeps tilt direction (1); Equator: about 12 hours (1); Poles: months of day and months of night (1); Seasons reversed in the two hemispheres (1).	1+1+1+1+1
	OR: Day: one rotation, noon to noon by shadow (1); Month: one cycle of Moon phases, about 29.5 days (1); Year: one revolution, seen from the seasons and stars (1); Shadow stick, Moon phases and star positions used (1); Calendars combine all three (1).	1+1+1+1+1
20	Solar: 365.25 days, leap year (1); Lunar: 354 days (1); Lunisolar: months follow the Moon (1); Extra month added (1); Festivals follow seasons (1).	1+1+1+1+1
	OR: Monsoon timing observed by constellations (1); Seasons tied to the Sun's position (1); Panchang used the Moon and Sun (1); Festivals marked agricultural events (1); Observations were passed across generations (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 11: **Keeping Time with the Skies** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** a full moon day	Earth between the Sun and the Moon.
2	**(c)** increases	Approaching the summer solstice.
3	**(d)** one day (24 hours roughly)	Earth's rotation.
4	**(a)** the whole night side of the Earth	Wide visibility.
5	**(c)** decreases	Moving toward winter.
6	**(d)** Earth revolves around the Sun	Earth moves about 1 degree per day in its orbit.
7	**(b)** a total solar eclipse	Region of total shadow.
8	**(d)** its time period of revolution around the Earth	Hence one face is seen.
9	**(b)** changes its position in its orbit	Different night direction.
10	**(a)** only with proper filters	Safety.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Follow lunar dates (1); Lunar and solar years differ by 11 days (1).	1+1
12	2000, 2004, 2008, 2012, 2016, 2020 (1); Six leap years (1).	1+1
13	Moon's orbit is tilted to the Earth's orbit (1); Alignment occurs only occasionally (1).	1+1
14	Samrat Yantra: tells time (1); Ram Yantra: measures altitude and azimuth of the Sun and stars (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Walk around a friend keeping your face to them (1); You make one turn in a full circuit (1); Moon's rotation equals revolution (1).	1+1+1
16	Lamp as the Sun, ball as the Moon (1); Move the ball around your head (1); See lit parts change as phases (1).	1+1+1
17	360 days, so 5.25 days short (1); Some months have 31 days (1); Leap year adds one day (1).	1+1+1
	OR: 0.25 day x 4 = 1 day (1); Add 29 February every four years (1); Century years not divisible by 400 are not leap (1).	1+1+1
18	Earth's axis is tilted (1); In summer the Sun is higher in the sky (1); In winter it is lower (1).	1+1+1
	OR: Earth's shadow on the Moon is large (1); Moon's shadow on the Earth is small (1); Hence solar eclipse seen from narrow path (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Solar: new moon, Moon in front of Sun (1); Lunar: full moon, Earth in between (1); Umbra and penumbra (1); Safe viewing with filters only (1); Eclipses occur only when the orbits line up (1).	1+1+1+1+1
	OR: Rotate globe for day and night (1); Tilt the axis and move around lamp for seasons (1); Move the ball around for phases (1); Align for eclipses (1); Observations compare with real sky (1).	1+1+1+1+1
20	Samrat Yantra tells time (1); Ram Yantra gives angles (1); Jai Prakash measures positions (1); Aryabhata: rotation and year (1); Used to prepare calendars (1).	1+1+1+1+1
	OR: Equator: day length 12 hours (1); North India: longer days in June (1); Noon Sun nearly overhead in June (1); Lower in December (1); Poles: extremes (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.